

Geometry
Unit 6
Lesson 11 Practice

Name Answer Key
Date _____
Period _____

$DKLM$ and $PTRS$ are shown, where $DKLM \sim PTRS$ with $MD = 21$, $DK = 4y - 2$, $KL = 6x + 3$, $LM = 36$, $PT = 10$, $TR = 11$, and $RS = 12$.

1. Determine the value of each variable.

$$x = 3$$

$$y = 4$$

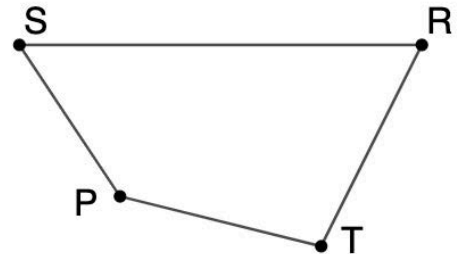
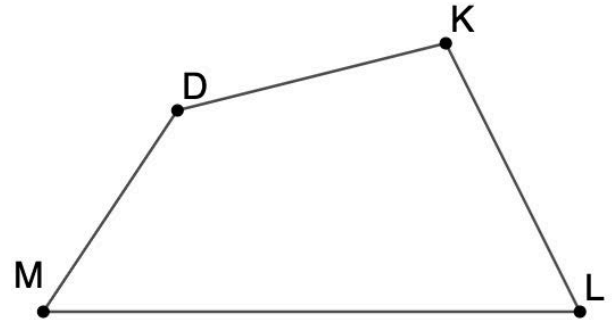
2. Determine the value of each side.

$$DK = 30$$

$$SP = 8$$

$$KL = 10$$

3. What is the perimeter of $DKLM$?



Consider additional measurements of angles for $DKLM$ and $PTRS$ where $m\angle MDK = (12x + 15)^\circ$, $m\angle SPT = (14x - 5)^\circ$, $m\angle DKL = 115^\circ$, $m\angle DML = (8y + 12)^\circ$, $m\angle PSR = (13y - 18)^\circ$.

4. Determine the value of each variable.

$$x = 10$$

$$y = 6$$

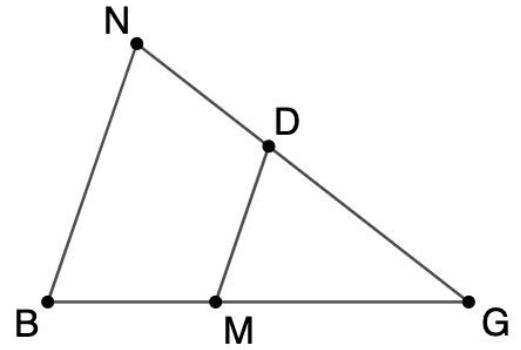
5. Determine the value of each angle.

$$m\angle MDK = 135^\circ$$

$$m\angle PTR = 115^\circ$$

$$m\angle PSR = 60^\circ$$

$\triangle NBG$ and $\triangle DMG$ are shown, where $\triangle NBG \sim \triangle DMG$ with $DM = 6$, $NB = 21$, $DG = 4$, $MG = 2x + 3$, and $BG = 13x - 1.5$.



6. What is the scale factor from $\triangle DMG$ to $\triangle NBG$?

$$\frac{21}{6} = 3.5$$

7. What is the value of the variable, x ?

$$3.5(2x + 3) = 13x - 1.5$$

$$x = 2$$

8. What is the length of BG ?

$$13(2) - 1.5 = 24.5 = BG$$

Consider additional measurements of angles for $\triangle NBG$ and $\triangle DMG$ where $m\angle GNB = (20y + 10)^\circ$, $m\angle GMD = (5x - 5)^\circ$, $m\angle GBN = (4x + 10)^\circ$, $m\angle GDM = (25y - 5)^\circ$.

9. Determine the value of each variable.

$$x = 15$$

$$y = 3$$

10. Determine the value of each angle.

$$m\angle GNB = 70^\circ$$

$$m\angle GBN = 70^\circ$$

$$m\angle DGM = 40^\circ$$